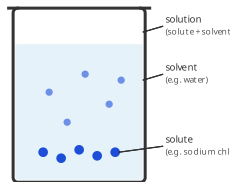


QUANTITATIVE CHEMISTRY 2 – CONCENTRATION OF SOLUTIONS

SOLUTION: A SOLUTE DISSOLVED IN A SOLVENT



The **solute** is the substance being dissolved.
The **solvent** is the liquid that does the dissolving.

CONCENTRATION FORMULA

$$\text{concentration (g/dm}^3\text{)} = \frac{\text{mass of solute (g)}}{\text{volume of solution (dm}^3\text{)}}$$

CONVERSION REMINDER

$$1000 \text{ cm}^3 = 1 \text{ dm}^3$$

WORKED EXAMPLE

A solution is made by dissolving 12 g of sodium chloride in 0.75 dm³ of water.

Calculate the concentration.

$$\begin{aligned} \text{concentration} &= \frac{\text{mass}}{\text{volume}} \\ &= \frac{12 \text{ g}}{0.75 \text{ dm}^3} \\ &= \mathbf{16 \text{ g/dm}^3} \end{aligned}$$

1

CONVERSION

Convert 250 cm³ to dm³.

(Use the conversion reminder above.)

Answer: _____ dm³

2

CALCULATE CONCENTRATION

A student dissolves 5.0 g of copper sulfate in 0.50 dm³ of water.

Calculate the concentration of the solution in g/dm³.

Answer: _____ g/dm³

3

CALCULATE MASS

A solution has a concentration of 24 g/dm³. The volume of the solution is 0.40 dm³.

Calculate the mass of solute in the solution.

Answer: _____ g

4

COMPARE SOLUTIONS

Two solutions are made.

Solution	Mass of solute (g)	Volume of solution (dm ³)
A	8.0	0.40
B	12.0	0.75

Which solution is **more concentrated**? Circle your answer.

A B

Show your working.